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Experiment no.-01

Vector Algebra / components and magnitude of vector

Qim:- Write a MATLAB program to solve the following examples.

Example :

$$\text{If } A = 100x - 40y + 60z$$

$$B = 20x + 0y$$

- find
- The component of A along ay .
 - The magnitude of $3A - B$.
 - A unit vector along A .

Apparatus : MATLAB software

Theory: A vector has both magnitude and direction.

The magnitude of A is a scalar written as $|A|$ or $|A|$. A unit vector a_A along A is defined as a vector whose magnitude is unity (i.e. 1) and its direction is along A i.e.

$$a_A = \frac{\vec{A}}{|\vec{A}|} = \frac{\vec{A}}{A}$$

① The components of A along ay is $A_y = -4$

$$\begin{aligned} \text{② } 3A - B &= 3(10, -4, -6) - (2, 1, 0) \\ &= (28, -13, -18) \end{aligned}$$



Hence,

$$|3A - B| = \sqrt{28^2 + (-13)^2 + (18)^2} = \sqrt{1297} \\ = 35.74$$

Two vectors A and B can be added together to give another vector C that is $C = A + B$.

The vector addition is carried out component by component thus if $A = (A_x, A_y, A_z)$ and $B = (B_x, B_y, B_z)$

$$C = (A_x + B_x)a_x + (A_y + B_y)a_y + (A_z + B_z)a_z.$$

Vector subtraction is similarly carried out as

$$D = A - B = A + (-B) \\ = (A_x - B_x)a_x + (A_y - B_y)a_y + (A_z - B_z)a_z.$$

Solution:

$$\textcircled{c} \text{ let } C = A + 2B \\ = (10, -4, 6) + (4, 2, 0) \\ = (14, -2, 6)$$

$$\text{A unit vector along } C \text{ is } a_c = \frac{C}{|C|} = \frac{(14, -2, 6)}{\sqrt{14^2 + (-2)^2 + 6^2}}.$$

or

$$a_c = 0.9113 a_x - 0.1302 a_y + 0.3906 a_z.$$

MATLAB code:

clc

clear

format ('r', 6)

A = [-10, -4, 6]

B = [2, 1, 0]

disp (A(1,2), 'component of A along ay:')

p = 3 * A - B

disp ((p(1,1)^2 + p(1,2)^2 + p(1,3)^2)^0.5, 'magnitude is:')

c = A + 2 * B.

det_c = (c(1,1)^2 + c(1,2)^2 + c(1,3)^2)^0.5.

format ('r', 7)

ac = c / det_c

disp (ac, 'unit vector along c is:')

output is-

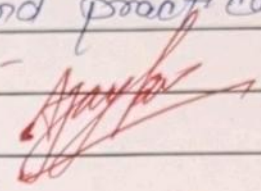
→ Component of A along ay:
-4

→ magnitude is:
35.74

→ unit vector along c is:
0.9113 - 0.1302 0.8906

Conclusion:

By doing this experiment, we come to know about the vector addition, subtraction and multiplication. Also here we found the theoretical and practical values are same.



Experiment no.-02

Vector Algebra

Aim: Write a MATLAB / SCILAB program to solve the following example.

ex- given vectors $A = 30x + 40y + 0z$ and $B = 20y - 50x$
Find the angle between A and B .

Apparatus: MATLAB / SCILAB software.

Theory: When two vectors A and B are multiplied, the result is either a scalar or a vector depending on how they are multiplied. Thus there are two types of vector multiplication

(i) Scalar (or dot) product: $A \cdot B$

(ii) Vector (or cross) product: $A \times B$

• The Dot product:

Given two vectors A and B , the dot product or scalar product is defined as the product of the magnitude of A , the magnitude of B and the cosine of the smaller angle between them

$$A \cdot B = |A| |B| \cos \theta_{AB}$$

The dot appears b/w the two vectors and should be made heavy for emphasis. The dot or scalar product is a scalar as one of names implies and it obeys the commutative law.

$$[A \cdot B = B \cdot A]$$

for the sign of the angle does not affect the cosine terms. The expression $A \cdot B$ is read as "A dot B".

* The cross product:

The vector multiplication of equation is called by "cross product" due to the cross sign. it is called vector product because the result is a vector if $A = (A_x \ A_y \ A_z)$ & $B = (B_x \ B_y \ B_z)$
thus

$$A \times B = \begin{vmatrix} a_x & a_y & a_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= (A_y B_z - A_z B_y) a_x + (A_z B_x - A_x B_z) a_y + (A_x B_y - A_y B_x) a_z$$

→ The angle θ_{AB} can be found by using either dot product (or cross product).

$$A \cdot B = (3, 4, 1) \cdot (0, 2, -5)$$

$$= 0 + 8 - 5 = 3$$

$$|A| = \sqrt{3^2 + 4^2 + 1^2} = \sqrt{26}, \quad |B| = \sqrt{0^2 + 2^2 + (-5)^2} = \sqrt{29}$$

$$\cos \theta_{AB} = \frac{A \cdot B}{|A||B|} = \frac{3}{\sqrt{26}\sqrt{29}} = 0.1092$$

$$\theta_{AB} = \cos^{-1}(0.1092)$$

$$\theta_{AB} = 83.73^\circ$$

$$A \times B = \begin{vmatrix} a_x & a_y & a_z \\ 3 & 4 & 1 \\ 0 & 2 & -5 \end{vmatrix}$$

$$= (-20 - 2)a_x + (0 + 15)a_y + (6 - 0)a_z$$

$$= (-22, 15, 6)$$

$$|A \times B| = \sqrt{(-22)^2 + (15)^2 + 6^2} = \sqrt{745}$$

$$\sin \theta_{AB} = \frac{|A \times B|}{|A||B|} = \frac{\sqrt{745}}{\sqrt{26}\sqrt{29}} = 0.994$$

$$\theta_{AB} = \cos^{-1}(0.994)$$

$$\theta_{AB} = 83.73^\circ$$

Procedure :

- i. Open MATLAB / SCILAB software
- ii. Goto Home and click on new script
- iii. Write MATLAB / SCILAB code
- iv. Save script file
- v. Click on Run Button
- vi. See the answer in command window
- vii. check output.

Program:

clc

clear

A = [3, 4, 1]

B = [0.2, -5]

det_A = (A(1,1)^2 + A(1,2)^2 + A(1,3)^2)^0.5

det_B = (B(1,1)^2 + B(1,2)^2 + B(1,3)^2)^0.5

theta = acosd(Lsum(A*B)) / (det_A * det_B)

disp(theta, 'Angle b/w A and B is')

Output -

→ Angle b/w A and B is : 83.727712.

Conclusion:

This experiment was completed successfully to determine the dot product and cross product of the given vector using MATLAB.

The results obtained from the program matched the theoretical calculation providing the accuracy of the method.

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25/10/2023

Experiment no. 03.

Transmission Lines / Finding various parameters.

Ques: Write a MATLAB / SCILAB program to solve the following Example.

Example - A distortionless line has $Z_0 = 60 \Omega$, $\alpha = 20 \text{ mNp/m}$, $u = 0.6c$ where c is the speed of light in a vacuum. Find R, L, G, C and λ at 100 MHz .

Apparatus : MATLAB / SCILAB software

Theory:

Distortionless line ($R/L = G/C$)

$$\frac{R}{L} = \frac{G}{C}$$

$$\gamma = \sqrt{RG \left(1 + \frac{j\omega L}{R}\right) \left(1 + \frac{j\omega C}{G}\right)}$$

$$= \sqrt{RG} \left(1 + \frac{j\omega L}{R}\right) = \alpha + j\beta$$

$$\alpha = \sqrt{RG}, \quad \beta = \omega \sqrt{LC}$$

$$Z_0 = \sqrt{\frac{R(1 + j\omega L/R)}{G(1 + j\omega C/G)}} = \sqrt{\frac{R}{G}} = \sqrt{\frac{L}{C}} = R_0 + jX_0$$



$$R_0 = \sqrt{\frac{R}{G}} = \sqrt{\frac{L}{C}} \quad X_0 = 0$$

$$\mu = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}} = f\lambda$$

Solution:

for distortionless line $RC = GL$ or $G = \frac{RC}{L}$
and hence

$$Z_0 = \sqrt{\frac{L}{C}}$$

$$G_0 = \sqrt{\frac{L}{C}}$$

$$(G_0)^2 = \frac{L}{C} \Rightarrow \frac{L}{C} = 3600 \quad \text{--- (i)}$$

$$v = 0.6c$$

$$[\because c = 3 \times 10^8 \text{ m/s}]$$

$$v = 0.6 \times 3 \times 10^8$$

$$\boxed{v = 1.8 \times 10^8}$$

$$\frac{1}{\sqrt{LC}} = 1.8 \times 10^8$$

$$1 = 1.8 \times 10^8 \times \sqrt{LC}$$

$$\left(\frac{1}{1.8 \times 10^8}\right)^2 = LC$$

$$LC = 0.3086 \times 10^{-16}$$

$$LC = 3.086 \times 10^{-17} \quad \text{--- (ii)}$$



Substitute the value of L in eqnⁿ (11)

$$C^2(3600) = 3.086 \times 10^{-17}$$

$$C^2 = \frac{3.086 \times 10^{-17}}{3600}$$

$$C = 9.2 \times 10^{-11} \text{ fm} \quad , \quad L = 3.33 \times 10^{-17} \text{ H/fm}$$

$$Z_0 = \sqrt{\frac{L}{C}}$$

$$Z_0 = \sqrt{\frac{R}{G} \times \frac{R}{R}}$$

$$Z_0 = \frac{R}{\sqrt{RG}} \Rightarrow Z_0 = \frac{R}{\alpha} \quad [\because \alpha = \sqrt{RG}]$$

$$R = 60 \times 0.020 \text{ N/m}$$

$$R = 0.12 \text{ N/m}$$

$$A = \frac{C}{f} = \frac{v}{f} = \frac{1.8 \times 10^8}{100 \times 10^6}$$

$$A = 1.8 \text{ m}$$



code :

clear

clc

$$Z_0 = 60, \alpha = 20 \times 10^{-3} + 45.6 \times 3 \times 10^8, f = 100 \times 10^6$$

$$R = \alpha \times Z_0, \text{disp}(R, 'R=')$$

$$L = Z_0 / \mu, \text{disp}(L \times 10^9, 'L \text{ in nH} =')$$

$$G = \alpha \times \alpha / R, \text{disp}(G \times 10^6, 'G \text{ in micro S per meter} =')$$

$$C = 1 / (4 \times Z_0), \text{disp}(C \times 10^{12}, 'C \text{ in pF} =')$$

$$1 = \mu / f; \text{disp}(1, '1 =')$$

Output: $Z_0 = 60$

$$\alpha = 0.02$$

$$\mu = 1.800 \times 10^8$$

$$R = 1.2$$

$$L = 0.0000003$$

$$L \text{ in nH} = 333.3333$$

$$G = 0.0003333$$

$$G \text{ in micro S per meter} = 333.3333$$

$$C = 9.259 \times 10^{-11}$$

$$C \text{ in pF} = 92.592593$$

$$L = 1.8$$

Conclusion: By using this experiment, we verified the various transmission parameters for distortionless line. Also we found the theoretical and practical values are same.

[Signature]
30/05/2020

Experiment no.-04. COULOMB'S LAW

Qim:- Write MATLAB code to solve the following examples.

Example: point charge 1 mC and -2 mC are located at $(3, 2, 1)$ and $(-1, -1, 4)$ respectively. Calculate the electric force on a 10 nC charge located at $(0, 3, 1)$ and the electric field intensity at that point.

Apparatus - MATLAB Software.

Theory:-

→ Coulomb's law states that the force F between two point charges Q_1 and Q_2 is

① Along the line joining them.

② Directly proportional to the product Q_1, Q_2 of the charges.

③ Inversely proportional to the square of the distance R between them.

$$F_{12} = \frac{Q_1 Q_2}{4\pi \epsilon_0 R^2} \hat{r}_{12}$$

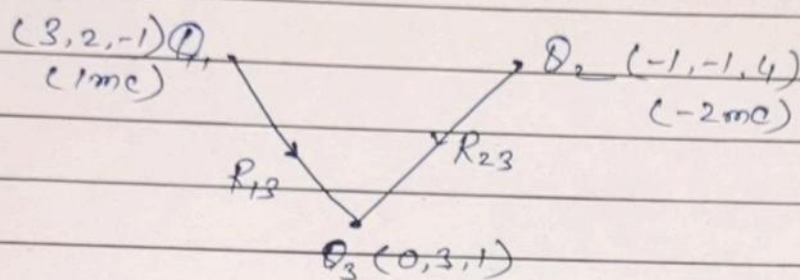
$$\epsilon_0 = 8.854 \times 10^{-12} \sim \frac{10^{-9}}{36\pi} \text{ F/m}$$

or

$$k = \frac{1}{4\pi \epsilon_0} \approx 9 \times 10^9 \text{ m/F}$$



→ The electric field intensity (or electric field strength).
 E is the force per unit charge when placed in the electric field.



$$\hat{R} = \frac{\vec{R}}{|\vec{R}|} \Rightarrow \hat{R}_{13} = \frac{\vec{R}_{13}}{|\vec{R}_{13}|}$$

$$\vec{F} = \frac{Q_1 Q_2}{4\pi \epsilon_0 R_{13}^2} \hat{R}_{13} + \frac{Q_2 Q_1}{4\pi \epsilon_0 R_{23}^2} \hat{R}_{23}$$

$$\vec{F} = \frac{Q_3}{4\pi \epsilon_0} \left[\frac{Q_1 \vec{R}_{13}}{R_{13}^2} + \frac{Q_2 \vec{R}_{23}}{R_{23}^2} \right]$$

$$\begin{aligned} \vec{R}_{13} &= (0, 3, 1) - (3, 2, -1) \\ &= (0\hat{x} + 3\hat{y} + \hat{z}) - (\hat{x} + 2\hat{y} - \hat{z}) \end{aligned}$$

$$\vec{R}_{13} = -\hat{x} + \hat{y} + 2\hat{z}$$

$$\begin{aligned} |\vec{R}_{13}| &= \sqrt{(-1)^2 + (1)^2 + (2)^2} \\ &= \sqrt{14} \end{aligned}$$

$$\begin{aligned} \vec{R}_{23} &= (0, 3, 1) - (-1, -1, 4) \\ &= (0\hat{x} + 3\hat{y} + \hat{z}) - (-\hat{x} - \hat{y} + 4\hat{z}) \end{aligned}$$



$$\vec{R}_{23} = \hat{a}_x + 4\hat{a}_y - 3\hat{a}_z$$

$$|\vec{R}_{23}| = \sqrt{1^2 + 4^2 + (-3)^2} = \sqrt{1 + 16 + 9} = \sqrt{26}$$

$$\vec{F} = 10 \times 10^{-9} \times 9 \times 10^4 \left[\frac{(1 \times 10^{-3}) \cdot (-3\hat{a}_x + \hat{a}_y + 2\hat{a}_z)}{14\sqrt{4}} - \frac{(2 \times 10^{-3}) \cdot (\hat{a}_x + \hat{a}_y - 3\hat{a}_z)}{26\sqrt{26}} \right]$$

$$\vec{F} = 10 \times 9 \times 10^{-3} \left[\frac{(-3\hat{a}_x + \hat{a}_y + 2\hat{a}_z)}{52.38} - \frac{(2\hat{a}_x + \hat{a}_y - 3\hat{a}_z)}{132.57} \right]$$

$$\vec{F} = 9 \times 10^{-2} [-0.057\hat{a}_x + 0.019\hat{a}_y + 0.038\hat{a}_z - 0.15\hat{a}_x - 0.06\hat{a}_y + 0.045\hat{a}_z]$$

$$= 9 \times 10^{-2} [-0.07\hat{a}_x - 0.041\hat{a}_y + 0.083\hat{a}_z]$$

$$= 0.648 \times 10^{-2} \hat{a}_x - 0.369 \times 10^{-2} \hat{a}_y + 0.74 \times 10^{-2} \hat{a}_z$$

$$F = 16.48 \hat{a}_x - 3.60 \hat{a}_y + 7.47 \hat{a}_z$$

$$\vec{E} = \frac{\vec{F}}{q} = \frac{\vec{F}}{10 \times 10^{-9}}$$

$$= (-6.48, -3.69, 4.77) \cdot \frac{10^{-3}}{10 \times 10^{-9}}$$

$$\vec{E} = -6.48 \hat{a}_x - 3.69 \hat{a}_y + 7.47 \hat{a}_z \text{ Kv/m.}$$

code :-

```
clear
```

```
clc
```

```
format ('v',6)
```

```
Q1 = 1
```

```
Q2 = -2
```

```
Q = 10*10^-9
```

```
P1 = [0 3 1] - [3 2 -1]
```

```
P2 = [0 3 1] - [-1 -1 4]
```

```
e = 10^-9 / (36*pi*P1)
```

```
def1 = (P1(1,1)^2 + P1(1,2)^2 + P1(1,3)^2)^.5
```

```
def2 = (P2(1,1)^2 + P2(1,2)^2 + P2(1,3)^2)^.5
```

```
F = [(Q*Q1)/(4*pi*e*(def1)^3) +  
      [(Q*Q2)/(4*pi*e*(def2)^3)]
```

```
E = (10^-6)*(F/Q)
```

```
disp(F, 'F(in mN) = ')
```

```
disp(E, 'At that point E (in kV) = ')
```

Output :

```
F (in mN) = -6.512 -38.713 7.509
```

```
At that point E (in kV) = -651.2 -371.3 750.9.
```

Conclusion :- By doing this experiment, we verified the coulomb's law. Also we found the theoretical & practical values are same.

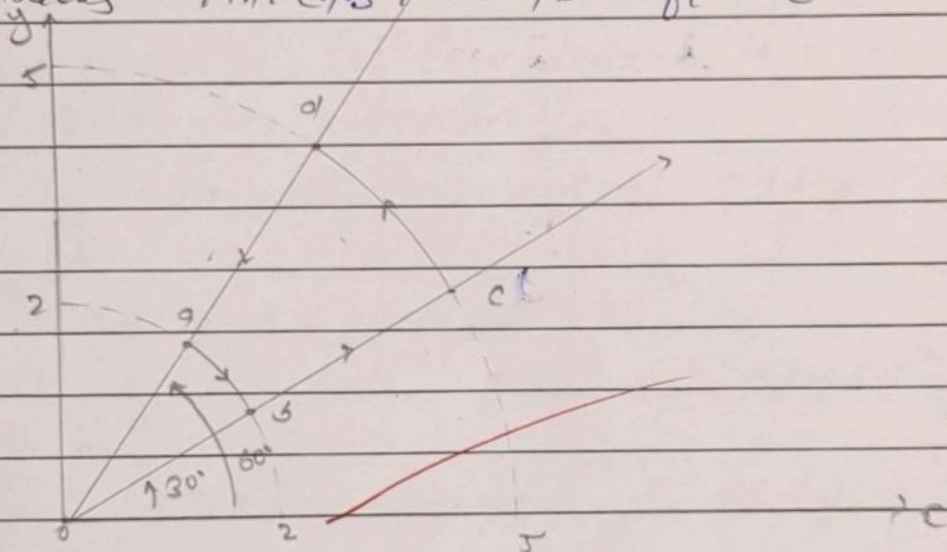
Experiment no.-05. Stoke's Theorem.

Aim:- Write a MATLAB /SCILAB program to solve the following Example.

Ex-

If $A = e \cos \phi \mathbf{a}_\rho + \sin \phi \mathbf{a}_\phi$, evaluate $\oint A \cdot d\mathbf{l}$ around the path shown in figure confirms this using Stoke's theorem.

Apparatus - MATLAB /SCILAB software.



Theory:- Stokes's theorem states that the circulation of a vector field A around a (closed) path " L " is equal to the surface integral of the curl of A over the open surface S bounded by L . provided that A and $\nabla \times A$ are continuous on S .



$$\oint_L \vec{A} \cdot d\vec{L} = \oint_S (\nabla \times \vec{A}) \cdot d\vec{S}$$

Solution:

$$\vec{A} = r \cos \phi \hat{r} + \sin \phi \hat{\phi}$$

$$\oint \vec{A} \cdot d\vec{L} = \oint_S (\nabla \times \vec{A}) \cdot d\vec{S}$$

$$\oint \vec{A} \cdot d\vec{L} = \left[\int_a^b \vec{A} \cdot d\vec{L} + \int_b^c \vec{A} \cdot d\vec{L} + \int_c^d \vec{A} \cdot d\vec{L} + \int_d^a \vec{A} \cdot d\vec{L} \right]$$

whose path "L" has been divided into segments ab, bc, cd & da as shown in fig.

Along ab, $r=2$ and $d\vec{L} = r d\phi \hat{\phi}$. Hence

$$\int_a^b \vec{A} \cdot d\vec{L} \Rightarrow \phi \Rightarrow 60^\circ \text{ to } 30^\circ$$

$$r=2, d\vec{L} = r d\phi \hat{\phi}$$

$$\Rightarrow \int_{60}^{30} [r \cos \phi \hat{r} + \sin \phi \hat{\phi}] \cdot r d\phi \hat{\phi} \quad \{ r=2 \}$$

$$\Rightarrow \int_{60}^{30} [r \cos \phi r d\phi \hat{r} \hat{\phi} + \sin \phi r d\phi \hat{\phi} \hat{\phi}]$$

$$\Rightarrow \int_{60}^{30} [r \sin \phi d\phi]_{r=2} = 2 \int_{60}^{30} \sin \phi d\phi \quad \left\{ \begin{array}{l} \hat{r} \cdot \hat{r} = 1 \\ \hat{\phi} \cdot \hat{\phi} = 1 \\ \hat{r} \cdot \hat{\phi} = 0 \end{array} \right.$$



$$= 2 \left[-\cos \phi \right]_{60}^{90}$$

$$= -(\sqrt{3} - 1).$$

* Along bc, $\phi = 30^\circ$ and $dl = de \hat{e}$. Hence

$$\int_b^c \vec{A} \cdot d\vec{l} \Rightarrow \phi = 30^\circ, \quad r = 2 \text{ to } 5$$

$$= \int_b^c [r \cos \phi \hat{e} + \sin \phi \hat{\phi}] dr \hat{e}$$

$$= \int_b^c [r \cos \phi dr] \Rightarrow \int_2^5 r \cos 30^\circ dr = \frac{\sqrt{3}}{2} \int_2^5 r dr$$

$$= \frac{\sqrt{3}}{4} [r^2]_2^5 = \frac{\sqrt{3}}{4} [25 - 4] = \frac{21\sqrt{3}}{4}.$$

* Along cd, $r = 5$ and $dl = r d\phi \hat{\phi}$. Hence.

$$\int_c^d \vec{A} \cdot d\vec{l} \Rightarrow \phi = 30^\circ \text{ to } 60^\circ, \quad r = 5$$

$$= \int_c^d [r \cos \phi \hat{e} + \sin \phi \hat{\phi}] (r d\phi \hat{\phi})$$

$$= \int_e^d [0 + r \sin \phi d\phi] = \int_{30^\circ}^{60^\circ} 5 \sin \phi d\phi$$

$$= 5 [-\cos \phi]_{30^\circ}^{60^\circ} = \frac{5}{2} (\sqrt{3} - 1)$$

* Along $d\phi$, $\phi = 60^\circ$ and $d\mathbf{l} = d\rho \hat{\mathbf{e}}_\rho$. Hence

$$\int_d \bar{\mathbf{A}} \cdot d\mathbf{l} \Rightarrow \rho = 5 \text{ to } 2, \phi = 60^\circ$$

$$= \int_d [\rho \cos \phi \hat{\mathbf{e}}_\rho + \sin \phi \hat{\mathbf{e}}_\phi] d\rho \hat{\mathbf{e}}_\rho$$

$$= \int_d [\rho \cos \phi d\rho] = \int_5^2 \rho \cos(60^\circ) d\rho$$

$$= \frac{1}{2} \int_5^2 \rho d\rho = \frac{1}{2} \left[\frac{\rho^2}{2} \right]_5^2 = \frac{1}{4} [4 - 25] = -\frac{21}{4}$$

Putting all these together result is.

$$\oint \bar{\mathbf{A}} \cdot d\mathbf{l} = -(\sqrt{3} - 1) + \frac{2\sqrt{3}}{4} + \frac{5}{2}(\sqrt{3} - 1) - \frac{21}{4} = 4.941$$

Using Stoke's theorem (because "L" is a closed path).

$$\oint \vec{A} \cdot d\vec{l} = \int_S (\nabla \times \vec{A}) \cdot d\vec{S}$$

$$\times \int_S (\nabla \times \vec{A}) \cdot d\vec{S} \Rightarrow$$

$$\left\{ \begin{array}{l} \vec{A} = e \cos \phi \hat{a}_e + \sin \phi \hat{a}_\phi + 0 \\ \vec{A} = A_e \hat{a}_e + A_\phi \hat{a}_\phi + A_z \hat{a}_z \end{array} \right.$$

$$\nabla \times \vec{A} = \frac{1}{e} \begin{vmatrix} \hat{a}_e & \hat{a}_\phi & \hat{a}_z \\ \frac{\partial}{\partial e} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_e & e A_\phi & A_z \end{vmatrix}$$

$$\begin{array}{l} A_e, A_\phi, A_z \\ A_e = e \cos \phi \\ A_\phi = \sin \phi \\ A_z = 0 \end{array}$$

$$= \frac{1}{e} \left[\left\{ \frac{\partial A_z}{\partial \phi} - \frac{\partial (e A_\phi)}{\partial z} \right\} \hat{a}_e - \left\{ \frac{\partial A_e}{\partial z} - \frac{\partial A_z}{\partial e} \right\} \hat{a}_\phi + \left\{ \frac{\partial (e A_\phi)}{\partial e} - \frac{\partial A_e}{\partial \phi} \right\} \hat{a}_z \right]$$

$$= \frac{1}{e} \left[(0-0) \hat{a}_e - (0-0) \hat{a}_\phi + \left\{ \sin \phi \frac{\partial (e)}{\partial e} - e \frac{\partial (\cos \phi)}{\partial \phi} \right\} \hat{a}_z \right]$$

$$\nabla \times \vec{A} = \frac{1}{e} [\sin \phi + e \sin \phi] = \frac{1}{e} (1+e) \sin \phi \hat{a}_z$$

$$= \int_S (\nabla \times \vec{A}) \cdot d\vec{S} \Rightarrow \int_S \frac{1}{e} (1+e) \sin \phi \hat{a}_z \cdot d\vec{S}$$

$$= \int \frac{1}{e} (1+e) \sin \phi \hat{a}_z \cdot e d\phi d\theta \hat{a}_z$$

$$= \int_{\phi=30^\circ}^{60^\circ} \int_{e=2}^5 (1+e) \sin \phi \, d\phi \, de \Rightarrow \int_{30^\circ}^{60^\circ} \sin \phi \, d\phi \int_2^5 (1+e) \, de$$

$$= \left[-\cos \phi \right]_{30}^{60} \cdot \left[e + \frac{e^2}{2} \right]_2^5$$

$$= 4.941.$$

Procedure -

- Open MATLAB/SCILAB software
- Go to Home and click on new script
- ~~See~~ Write MATLAB/SCILAB code
- Save Script file
- Click on Run Button.
- See the theorem in command window
- Check results.

Program:-

```
clear
clc

ab = integrate ('2^x sin(xP)', 'P', %pi/3, %pi (6);
bc = (3^5/2)^x integrate ('P', 'P', 2, 5)
cd = integrate ('5^x sin(xP)', 'P', %pi/6, %pi/3)
da = .5^x integrate ('P', 'P', 5, 2)
c1 = ab + bc + cd + da.
```

```
disp (C1, 'C1 =')
```

```
C2 = integrate ('sin (phi)', 'phi', %pi/6, %pi/3) * integrate  
(' (1+p)', 2, 5).
```

```
disp (C2, 'C2 =')
```

```
disp ('Since C1 = C2 hence stoke's theorem is proved')
```

Output -

$$C_1 = 4.941343$$

$$C_2 = 4.941343$$

Since $C_1 = C_2$ hence stoke's theorem is proved.

Conclusion - By doing this Experiment, we verified the stoke's theorem. we found the theoretical & practical values are same.

Signature
14/07/2020



Experiment no. - 06

COORDINATE SYSTEMS AND TRANSFORMATION.

Aim:- Write a MATLAB program to solve the following example.

Apparatus:- MATLAB / SCILAB software.

Example - Express vector $\vec{r} = \frac{10}{\sigma} \hat{a}_r + r \cos \theta \hat{a}_\theta + r \hat{a}_\phi$ in cartesian and cylindrical coordinates. Find $B(-3, 4, 0)$ and $B(5, \pi/2, -2)$.

Sol:- we have

$$\vec{r} = \frac{10}{\sigma} \hat{a}_r + r \cos \theta \hat{a}_\theta + r \hat{a}_\phi$$

$$\text{Here, } r_x = \frac{10}{\sigma}, \quad r_\theta = r, \quad r_\phi = 1$$

$$\begin{bmatrix} r_x \\ r_y \\ r_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} \frac{10}{\sigma} \\ r \cos \theta \\ 1 \end{bmatrix}$$

$$r_x = \frac{10}{\sigma} \sin \theta \cos \phi + r \cos^2 \theta \cos \phi - \sin \phi \quad \text{--- (i)}$$

$$r_y = \frac{10}{\sigma} \sin \theta \sin \phi + r \cos^2 \theta \sin \phi + \cos \phi \quad \text{--- (ii)}$$

$$r_z = \frac{10}{\sigma} \cos \theta - r \cos \theta \sin \theta$$



$$\beta(x, y, z) = \beta(-3, 4, 0)$$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{9 + 16 + 0} = \sqrt{25} = 5$$

then,

$$i) z = r \cos \theta \Rightarrow \cos \theta = \frac{z}{r} = \frac{0}{5} = 0$$

$$\therefore \cos \theta = 0$$

$$ii) y = r \sin \phi \Rightarrow \sin \phi = \frac{y}{r} = \frac{4}{\sqrt{x^2 + y^2}} \quad [\because r = \sqrt{x^2 + y^2}]$$

$$\therefore \sin \phi = \frac{4}{\sqrt{25}} = \frac{4}{5}$$

$$iii) r \cos \phi = x$$

$$\therefore \cos \phi = \frac{x}{\sqrt{x^2 + y^2}} = -\frac{3}{5}$$

$$iv) y = r \sin \theta \sin \phi$$

$$\therefore \sin \theta = \frac{y}{r \sin \phi} = \frac{4}{5 \times \frac{4}{5}} = 1$$

then from (i)

$$\begin{aligned} \Delta x &= \frac{10}{5} \sin \theta \cos \phi + r \cos^2 \theta \cos \phi - \sin \phi \\ &= 2 \times 1 \times \left(-\frac{3}{5}\right) + 5 \times 0 \times \left(-\frac{3}{5}\right) - \frac{4}{5} \\ &= -\frac{6}{5} - \frac{4}{5} = -\frac{10}{5} = -2 \end{aligned}$$

from (ii)

$$\begin{aligned} \Delta y &= \frac{10}{5} \sin \theta \sin \phi + r \cos^2 \theta \sin \phi + \cos \phi \\ &= 2 \times 1 \times \frac{4}{5} + 0 + \left(-\frac{3}{5}\right) \\ &= \frac{8}{5} - \frac{3}{5} = 1 \end{aligned}$$

from (i)

$$A_z = \frac{10}{8} \cos \theta - \pi \cos \theta \sin \phi$$

$$= 2 \times 0 - \pi \times 0 \times \frac{4}{5}$$

$$= 0$$

$$\therefore \vec{A} = -2\hat{x} + \hat{y} \quad (\text{vector in cartesian form}).$$

(ii) Vector in cylindrical form.
we have

$$A(r, \phi, z) = (5, \pi/2, -2)$$

$$\text{and } \vec{A} = \frac{10}{8} \hat{r} + \pi \cos \theta \hat{\theta} + \hat{\phi}.$$

$$\text{here } A_r = \frac{10}{8}, A_\theta = \pi \cos \theta, A_\phi = 1.$$

$$\begin{bmatrix} B_r \\ B_\theta \\ B_z \end{bmatrix} = \begin{bmatrix} \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \times \begin{bmatrix} \frac{10}{8} \\ \pi \cos \theta \\ 1 \end{bmatrix}$$

$$B_r = \frac{10}{8} \sin \theta + \pi \cos^2 \theta \quad \text{--- (iv)}$$

$$B_\theta = 1 \quad \text{--- (v)}$$

$$B_z = \frac{10}{8} \cos \theta - \pi \sin \theta \cos \theta \quad \text{--- (vi)}$$



as we know

$$r = \sqrt{x^2 + y^2 + z^2} \quad \text{and} \quad \rho = \sqrt{x^2 + y^2}$$

$$r^2 = x^2 + y^2 + z^2$$

$$r = \sqrt{\rho^2 + z^2}$$

$$= \sqrt{25 + 4} = \sqrt{29}$$

then

$$i) z = r \cos \theta$$

$$\therefore \cos \theta = \frac{z}{r} = \frac{-2}{\sqrt{29}}$$

$$ii) \rho = r \sin \theta$$

$$\sin \theta = \frac{\rho}{r} = \frac{5}{\sqrt{29}}$$

then from (iv)

$$\begin{aligned} B_\rho &= \frac{10}{\rho} \sin \theta + r \cos \theta = \frac{10}{\sqrt{29}} \times \frac{5}{\sqrt{29}} + \sqrt{29} \times \frac{-2}{\sqrt{29}} \\ &= \frac{50 - 20\sqrt{29}}{\sqrt{29}} = 2.467 \end{aligned}$$

$$B_\phi = 1 \quad \text{from (v)}$$

from (vi)

$$B_z = \frac{10}{\sqrt{29}} \times \frac{(-2)}{\sqrt{29}} - \sqrt{29} \times \frac{(-2)}{\sqrt{29}} \times \frac{5}{\sqrt{29}} = \frac{20 + 10\sqrt{29}}{29}$$

$$B_z = 1.167$$

$$\vec{B} = 2.467 \hat{\rho} + 1 \hat{\phi} + 1.167 \hat{z}$$



Procedure :-

- Open MATLAB software
- Go to home and click on new script file
- Write MATLAB code
- Save script file
- Click on Run button
- See the Answer in command window
- Check results.

MATLAB code :-

```

clc
clear all
close all
format
[x,y,z] = sptocart (-3,4,0)
disp ('B in cartesian co-ordinate :')
disp ([x y z])
[r, p, z] = sptocyl (5, Pi/2, -2)
disp ([r p z])
function [x,y,z] = sptocart (x,y,z)
R = sqrt (x^2 + y^2)
r = sqrt (x^2 + y^2)
p = a sin (x/R)
Q = a cos (x/R)
X = (10/R) * sin (P) * cos (Q) + R * (cos (P))^2 * cos (Q)
    - sin (Q) ;
    
```



$$y = (1/R) * \sin(P) * \sin(Q) + R * (\cos(P))^2 * \sin(Q) + \cos(Q)$$

$$z = (1/R) * \cos(P) - R * \cos(P) * \sin(P)$$

end

function (r, p, z) = sptocylind(r, p, z)

$$R = \sqrt{r^2 + z^2}$$

$$P = \arccos(z/R)$$

$$r = (1/R) * \sin(P) + R * (\cos(P))^2$$

$$p = P$$

$$z = (1/R) * \cos(P) - R * \cos(P) * \sin(P)$$

end.

Output:

B in cartesian coordinates : -2 1 -1.937e-16

B in cylindrical coordinates : 2.4669 1.5708 1.1693.

Conclusion:- The experiment provided a clear understanding of coordinate systems and coordinate transformation. The comparison of theoretical & experimental results showed only negligible difference, validating the accuracy of the experiment.

[Signature]
28/11/20



Experiment no. 07.

DIFFERENT COMPONENTS OF A VECTOR.

Ques:- Write a MATLAB code to solve the following example.

Example - Given a vector field

$$\vec{D} = \sin\phi \vec{a}_r - \frac{1}{r} \sin\theta \cos\phi \vec{a}_\theta + r^2 \vec{a}_\phi$$

determine

- $\vec{D}$ at $P(10, 150^\circ, 330^\circ)$
- The component of $\vec{D}$ tangential to the spherical surface $r = 10$ at P .
- A unit vector at P perpendicular to $\vec{D}$ and tangential to the cone $\theta = 150^\circ$.

Apparatus : MATLAB / SCILAB software.

Solution:-

(a) At P , $r = 10$, $\theta = 150^\circ$, and $\phi = 330^\circ$. Hence,

$$\begin{aligned} \vec{D} &= 10 \sin 330^\circ \vec{a}_r - \frac{1}{10} \sin 150^\circ \cos 330^\circ \vec{a}_\theta + 100 \vec{a}_\phi \\ &= (-5, 0.043, 100) \end{aligned}$$

(b) Any vector $\vec{D}$ can always be resolved into two orthogonal components.

$$\vec{D} = \vec{D}_t + \vec{D}_n.$$



where D_t is tangential to a given surface and D_n is normal to it. In our case, since a_r is normal to the surface $r=10$

$$D_n = r \sin \phi \, a_r = -50a_r$$

Hence,

$$D_t = D - D_n = 0.043a_\theta + 100a_\phi.$$

c) A vector a perpendicular to D and tangential to the cone $\theta = 150^\circ$ is the same as the vector perpendicular to the both D and a_θ . Hence

$$D \times a_\theta = \begin{vmatrix} a_r & a_\theta & a_\phi \\ -5 & 0.043 & 100 \\ 0 & 1 & 1 \end{vmatrix}$$

$$= -100a_r - 5a_\phi$$

A unit vector along this is

$$a = \frac{-100a_r - 5a_\phi}{\sqrt{100^2 + 5^2}} = -0.9988a_r - 0.0499a_\phi.$$



Procedure :-

- Open MATLAB software
- Go to Home and click on new script file
- Write MATLAB code
- Save script file
- Click on the Run Button
- See the answer in command window
- Check results.

MATLAB code:

clc

clear all

close all

format ('v', 6)

function [R, P, Q] = Posvec (r, P, q).

$R = r * \sin(q)$; $P = -\sin(P) * \cos(q) / r$; $Q = r * r$;

$D = [R \ P \ Q]$

disp (D, 'Dat P');

$D_n = [r * \sin(q) \ 0 \ 0]$;

$D_t = D - D_n$;

disp (D_t, 'Tangential component of D at P');

End function

Posvec (10, 150, 330)

$D = [-5.043 \ 100]$;

$q = [0 \ 10]$

$$U_1 = D(1,2) * a(1,3) - D(1,3) * a(1,2)$$

$$U_2 = D(1,3) * a(1,1) - D(1,1) * a(1,3)$$

$$U_3 = D(1,1) * a(1,2) - D(1,2) * a(1,1)$$

$$U = [U_1 \quad U_2 \quad U_3]$$

$$\text{det}_U = \text{sqrt}(U_1^2 + U_2^2 + U_3^2);$$

format('v',7):

disp(U/det_U, 'Unit vector P perpendicular to D');

Output :-

D at P

-5 -0.043 100

Tangential component of D at P

0 -0.043 100

Unit vector P perpendicular to D

-0.9988 0 -0.0499

Conclusion: By doing this experiment, we come to know about different components of a vector.

Also we found the theoretical and practical values are same.

[Signature]
28/07/2020

Experiment no.-08

DIPOLE

Ques: Write a MATLAB / SCILAB program to solve the following example.

Example: Two dipoles with dipole moments $-50\hat{z}$ nC/m and $90\hat{z}$ nC/m are located at points $(0,0,-2)$ and $(0,0,3)$ respectively. Find the potential at the origin.

Apparatus: MATLAB / SCILAB software

Theory: An electric dipole is formed when two point charges of equal magnitude but opposite sign are separated by a small distance.

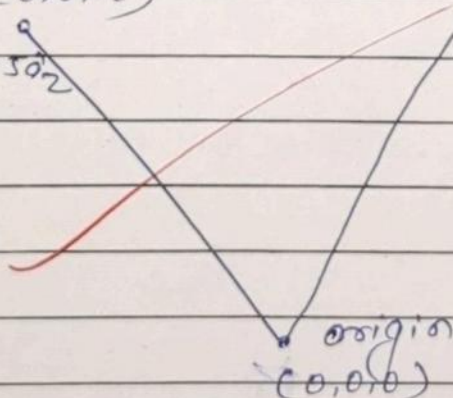
$$V(r) = \frac{P \cdot (r - r')}{4\pi \epsilon_0 |r - r'|^3}$$

Solution: $(0,0,-2)$

$$P_1 = -50\hat{z}$$

$(0,0,3)$

$$P_2 = 90\hat{z}$$





$$V = \frac{P_1 \vec{r}_1}{4\pi\epsilon_0 r_1^3} + \frac{P_2 \vec{r}_2}{4\pi\epsilon_0 r_2^3}$$

Here, $\vec{r}_1 = (0, 0, 0) - (0, 0, 2)$
 $= -2\hat{a}_z$

$$|\vec{r}_1| = 2$$

~~$\vec{r}_2 = (0, 0, 0) - (0, 0, 3)$~~
 ~~$= -3\hat{a}_z$~~

~~$$|\vec{r}_2| = 3$$~~

~~$$V = \left(\frac{-5\hat{a}_z \cdot 2\hat{a}_z}{4\pi\epsilon_0 2^3} + \frac{9\hat{a}_z \cdot (-3\hat{a}_z)}{4\pi\epsilon_0 3^3} \right) \times 10^{-9}$$~~

~~$$V = \left(\frac{-10}{4\pi 8\pi\epsilon_0} - \frac{27}{4\pi 27\pi\epsilon_0} \right) \times 10^{-9}$$~~

~~$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{-10}{8} - 1 \right) \times 10^{-9}$$~~

~~$$V = \frac{-18}{8} \times 9 \times 10^{-9} \times 10^{-9} = \frac{-18}{8} \times 9 = -20.25 \text{ V.}$$~~

Procedure :-

- Open MATLAB / SCILAB software
- Go to Home and click on new script file
- Write MATLAB code
- Save Script file
- Click on Run Button.

- see the Answer in command window
- Check results.

MATLAB code :-

clear

clc

$$P1 = -5 * 10^{-9}, P2 = 9 * 10^{-9}$$

$$r1 = 2, r2 = 3, e = 10^{-9} / (36 * \%pi);$$

$$V = (1 / (4 * \%pi * e)) * ((P1 * abs(r1) / r1^3) + (P2 * abs(r2) / r2^3));$$

disp(V);

Output :-

$$P1 = -5.0000 \times 10^{-9}$$

$$r1 = 2$$

$$r2 = -3$$

$$V = -20.25$$

Conclusion: By doing this experiment, we verified the dipole theory. Also we found the theoretical and practical values are same.

Signature
04/10/2026



Experiment no. - 09

Electromagnetic wave propagation / waves in dielectric.

Qim:- Write a MATLAB program to solve the following example.

Ex- In a lossless medium for which $\eta = 60\pi$, $\mu = 1$ and $H = -0.1 \cos(\omega t - z) \hat{a}_x + 0.5 \sin(\omega t - z) \hat{a}_y$ A/m, calculate E and ω .

Apparatus - MATLAB software.

Theory:-

In a lossless dielectric, $\sigma \ll \omega\epsilon$. It is a special case of that in Section 1.

$$\sigma = 0, \quad \epsilon = \epsilon_0 \epsilon_r, \quad \mu = \mu_0 \mu_r.$$

Substituting these into eqs.

$$\alpha = 0, \quad \beta = \omega \sqrt{\mu\epsilon}$$

$$\eta = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu\epsilon}}, \quad \lambda = \frac{2\pi}{\beta}$$

Also

$$\eta = \sqrt{\frac{\mu}{\epsilon}} \angle 0^\circ.$$

and thus E and H are in time phase with each other



Solution -

In this case, $\sigma = 0$, $\alpha = 0$ and $\beta = 1$ so.

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0}{\epsilon_0}} \sqrt{\frac{\mu_r}{\epsilon_r}} = \frac{120\pi}{\sqrt{\epsilon_r}}$$

$$\sqrt{\epsilon_r} = \frac{120\pi}{\eta} = \frac{120\pi}{60\pi} = 2 \Rightarrow \boxed{\epsilon_r = 4}$$

$$\beta = \omega \sqrt{\mu \epsilon} = \omega \sqrt{\mu_0 \epsilon_0} \sqrt{\mu_r \epsilon_r} = \frac{\omega}{c} \sqrt{4} = \frac{2\omega}{c}$$

$$\omega = \frac{\beta c}{2} = \frac{3 \times 10^8}{2} = 1.5 \times 10^8 \text{ rad/s.}$$

$$\boxed{\omega = 1.5 \times 10^8 \text{ rad/s}}$$

Procedure:

1. Open MATLAB software
2. Go to Home and click on New Script file.
3. Write MATLAB code
4. Save Script file
5. Click on Run Button
6. See the Answer in command window.
7. Check results.

Code:

clear

clc

B=1

$\eta = 60 \times \pi$

$U_0 = 1$

$E_0 = 10^{-9} / (36 \times \pi)$

$U_0 = 4 \times \pi \times 10^{-7}$

$E_0 = (U_0 \times U_0) / (\eta^2 \times E_0)$

~~fprint('E₀ = % of 1n', E₀)~~

$\omega = B / \sqrt{E_0 \times E_0 \times U_0 \times U_0}$

~~fprint('ω in mod 1sec = % of 1n'~~

$\omega = 10^{-6}$

Output:

$E_0 = 4$

$\omega = 100$ (mod 1sec)

Conclusion: By doing this experiment, we verified the theory of electromagnetic wave propagation of waves in dielectric. Also we found the theoretical and practical values are same.

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11/08/16



Experiment no. 10

Waveguide / Wave Propagation in Guide

Aim:- Write a MATLAB program to solve the following example.

Example. A standard air filled rectangular waveguide with dimensions $a = 8.636 \text{ cm}$, $b = 4.318 \text{ cm}$ is fed by a 4 GHz carrier from a coaxial cable. Determine if a TE_{10} mode will be propagated. If so, calculate the phase velocity and the group velocity.

Apparatus - MATLAB software.

Solution:

for the TE_{10} mode $f_c = c'/2a$. Since the waveguide is air filled $c' = c = 3 \times 10^8 \text{ m/s}$.

Hence,

$$f_c = \frac{3 \times 10^8}{2 \times 8.636 \times 10^{-2}} = 1.737 \text{ GHz}$$

As $f = 4 \text{ GHz} > f_c$, the TE_{10} mode will propagate.

$$v_p = \frac{c'}{\sqrt{1 - (f_c/f)^2}} = \frac{3 \times 10^8}{\sqrt{1 - (1.737/4)^2}} = 3.33 \times 10^8 \text{ m/s}$$



$$U_g = \frac{U^2}{U_p} = \frac{4 \times 10^{16}}{3.33 \times 10^8} = 2.702 \times 10^8 \text{ m/s}$$

Procedure:

1. Open MATLAB software
2. Go to Home and click on New Script file
3. Write MATLAB code.
4. Save Script file
5. Click on Run Button
6. See the Answer in Command window
7. Check results.

MATLAB code -

clear

clc

$a = 8.636 \times 10^{-2}$, $b = 4.318 \times 10^{-2}$, $f = 4 \times 10^9$

$u = 3 \times 10^8$

$f_c = u / (2 \times a)$

disp($f_c \times 10^{-9}$, 'cut off frequency =')

if ($f > f_c$) then disp('As $f > f_c$ so TE10 mode will propagate')

else disp('It will not propagate')

end

$U_p = u / \sqrt{1 - (f_c / f)^2}$

disp($U_p \times 10^{-6}$, 'Phase velocity in mm/sec =')

$U_g = u \times u / U_p$



disp ($V_g \approx 10^{-6}$, 'Group velocity in mm/sec =')

Output.

$$a = 0.08636$$

$$b = 0.04318$$

$$\text{Cut off frequency} = 1.7369152$$

As $f > f_c$ so TE₁₀ mode will propagate

$$\text{Phase velocity in mm/sec} = 333.0364$$

$$\text{Group velocity in mm/sec} = 272.24078$$

Conclusion: By doing this experiment, we verified the theory of waveguide & wave propagation in guide. Also we found the theoretical and practical values are same.

[Signature]
25/6/16